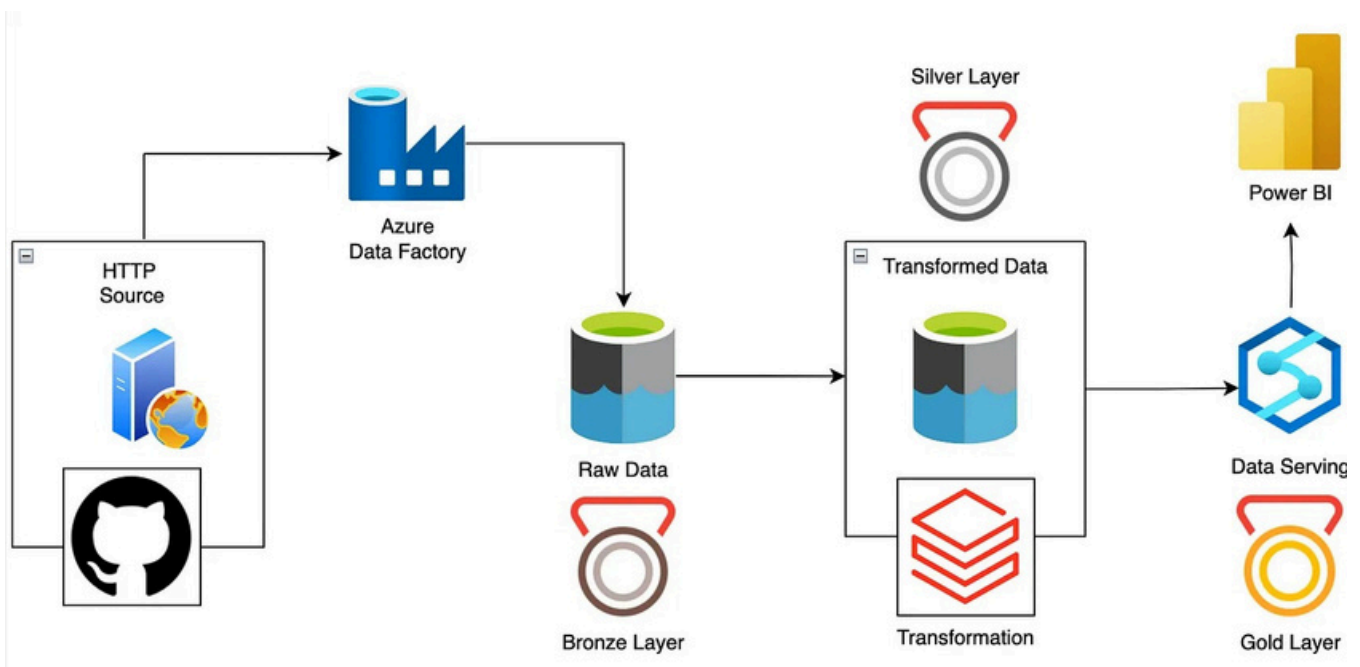


Azure End-To-End Data Engineering Project



This is an end-to-end Azure Data Engineering Project where I upload data from GitHub HTTP source to bronze layer through Azure Data factory. Transform the data in the silver layer using Databricks and provide the data to downstream (Analytics, Data Science) using Azure Synapse Analytics.

I have followed Medallion architecture where:

Raw data is uploaded to bronze layer using parameters and dynamic pipelines.

Raw data is Transformed in the solver layer using Databricks.

- Finally, the Transformed data is served into gold layer, where views of the data are created for downstream analysis.

ABOUT THE DATASET:

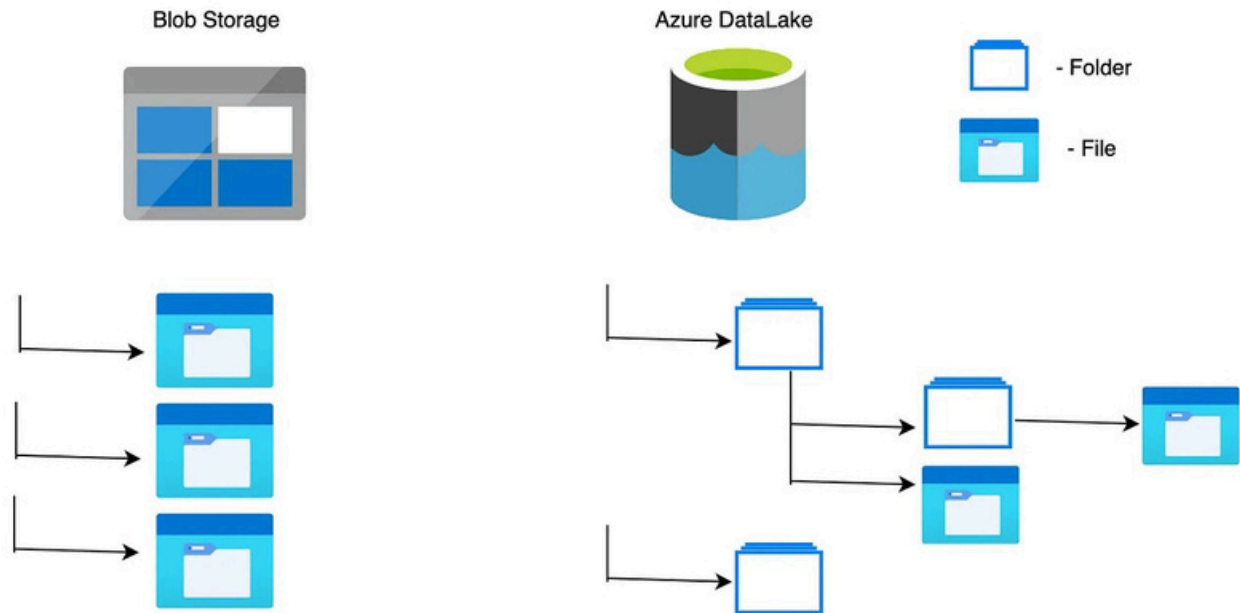
This is an open-source dataset from Kaggle named — Adventure Works.

Here is a quick description of the columns in the dataset.

- AdventureWorks_Calendar.csv
- AdventureWorks_Customers.csv
- AdventureWorks_Product_Categories.csv
- AdventureWorks_Product_Subcategories.cs
- v AdventureWorks_Products.csv
- AdventureWorks_Returns.csv
- AdventureWorks_Sales_2015.csv
- AdventureWorks_Sales_2016.csv
- AdventureWorks_Sales_2017.csv
- AdventureWorks_Territories.csv

I have used Azure DataLake Storage to store the data.

Why ADLS Gen 2?



Storage in Blog Vs. ADLS

In ADLS, we can store data in the form of hierarchies. While creating the storage account, make sure to enable “Hierarchical Namespace” to create ADLS, else a blob storage will be created by default.

Home > azureproject2025datalake

azureproject2025datalake | Containers ⚙️ ☆ ...

Storage account

Search

+ Container 🔒 Change access level ↶ Restore containers ↻ Refresh 🗑️ Delete 🔍

Search containers by prefix

Name	Last modified
☐ \$logs	14/01/2025, 17:59:41
☐ bronze	14/01/2025, 18:57:49
☐ gold	14/01/2025, 18:57:59
☐ silver	14/01/2025, 18:57:54

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Partner solutions
Data storage
Containers

Containers in Storage Account

I have created 3 containers to follow Medallion architecture (bronze, silver and gold)

THE BRONZE LAYER:

Now, I will load the data from GitHub to the bronze layer (raw data). I am first creating a static loading of data from GitHub to bronze, then I will create dynamic pulling of all files from a folder to bronze.

New linked service

 HTTP [Learn more](#) 

Name *

HttpGit

Description

Connect via integration runtime * 

 AutoResolveIntegrationRuntime 

Base URL *

https://raw.githubusercontent.com/

 Information will be sent to the URL specified. Please ensure you trust the URL entered.

Server certificate validation 

☒ Enable ☐ Disable

Authentication type * 

Anonymous 

Auth headers 

 New

Annotations

 New

> Parameters

> Advanced 

Create

Back

 Test connection

Cancel

HTTP Linked Service

I have created an HTTP Linked Service by giving the base URL of products.csv file from GitHub's Raw data.

Now, I will create a Azure DataLake Storage Linked Service to connect to ADLS.

New linked service

 Azure Data Lake Storage Gen2 [Learn more](#) 

Name *

AzureDataLakeService

Description

Connect via integration runtime * ⓘ



AutoResolveIntegrationRuntime



Authentication type

Account key



Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

Free Trial (e57a8b15-eab6-4488-8ade-b17cbb1ac1ab)



Storage account name *

azureproject2025datalake



Test connection ⓘ

☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

> Advanced ⓘ

Create

Back



Test connection

Cancel

ADLS Linked Service

COPY ACTIVITY:

Copy Activity requires the following information:

- Source: It requires the name of the dataset, linked service and relative path of the source dataset. (For data stored in ADLS, for GitHub we will be using an HTTP Linked Service)
- Sink: It requires the name of the dataset, linked service and relative path of the source dataset. (To store data in ADLS)

Set properties

Name

source_git

Linked service *

HttpGit

Relative URL

anshlamba03/Adventure-Works-Data-Engineering-Project/refs/heads/main/Data/Adven

First row as header



Import schema

☐ From connection/store ☐ From sample file ☒ None

> Advanced

Source Dataset from git

Set properties

Name

Linked service *





File path

/ /

 | 

First row as header ☒

Import schema

☐ From connection/store

☐ From sample file

☒ None

> Advanced

Sink Dataset for ADLS

After connecting Copy Activity to Source and Sink, now I will run the debug option to execute this Pipeline.

Pipeline run ID: f4782b99-d26c-4a02-9681-edcee7f95466   **Pipeline status**  Succeeded [View de](#)

All status 

[Monitor in Azure Metrics](#) 

Showing 1 - 1 of 1 items

Activity name 	Activity status 	Activity type 	Run start 	Duration 
CopyRawData	 Succeeded	Copy data	1/14/2025, 7:15:52 PM	13s

Successful Execution of Pipeline

Home > azureproject2025datalake | Containers >

bronze ...
Container

Search

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: bronze / csvfiles / anshlamba03 / Adventure-Works-Data-Engineering-Project / refs / heads / main / Data

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status
[..]			
AdventureWorks_Products.csv	14/01/2025, 19:16:04	Hot (Inferred)	

Data Successfully Copied to the bronze folder

Now, I will be creating a Copy Activity which will be parameterised.

I created a new Pipeline where I dragged a Copy Activity.

The source will be parameterised in the following way:

General Source Sink¹ Mapping Settings User properties

Source dataset * dynamic_copy_source Open + New Preview data Learn more

Dataset properties ⓘ

Name	Value	Type
p_rel_url	Value	string

Parameterised Source Dataset

The input (each rel_url will be coming from a ForEach Activity) of the Copy Activity receives a p_rel_url which will dynamically change upon each Copy step.

The sink will be parameterised in the following way:

General Source Sink Mapping Settings User properties

Sink dataset * dynamic_copy_sink Open + New Learn more

Dataset properties ⓘ

Name	Value	Type
p_folder_name	Value	String
p_file_name	Value	String

Parameterised Sink Dataset

I will now create a json file which will contain the following fields:

- p_rel_url
- p_sink_folder
- p_sink_file

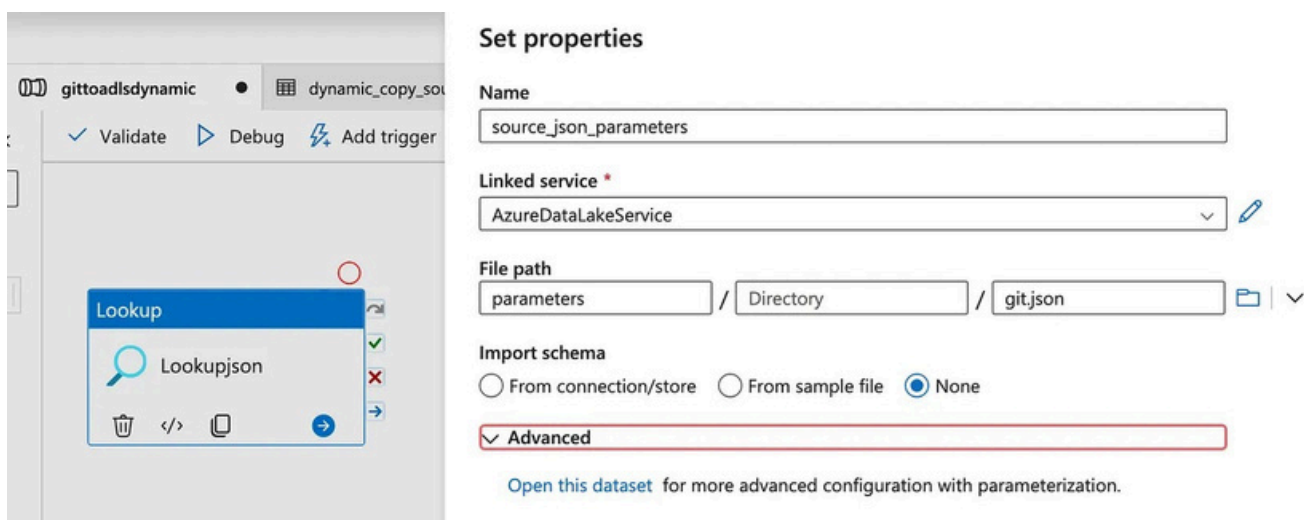
Sample json file snippet:

```
[
  {
    "p_rel_url" : "anshlamba03/Adventure-Works-Data-Engineering-Project/main/Data/AdventureWorks_Product_Categories.csv",
    "p_sink_folder" : "AdventureWorks_Product_Categories",
    "p_sink_file" : "AdventureWorks_Product_Categories.csv"
  },
  {
    "p_rel_url" : "anshlamba03/Adventure-Works-Data-Engineering-Project/main/Data/AdventureWorks_Calendar.csv",
    "p_sink_folder" : "AdventureWorks_Calendar",
    "p_sink_file" : "AdventureWorks_Calendar.csv"
  },
]
```

json file with Parameters

This git.json will be uploaded in the StorageLake under parameters folder.

I will be creating an Activity called LookUp (To get the information inside git.json file)



LookUp Activity to Read git.json

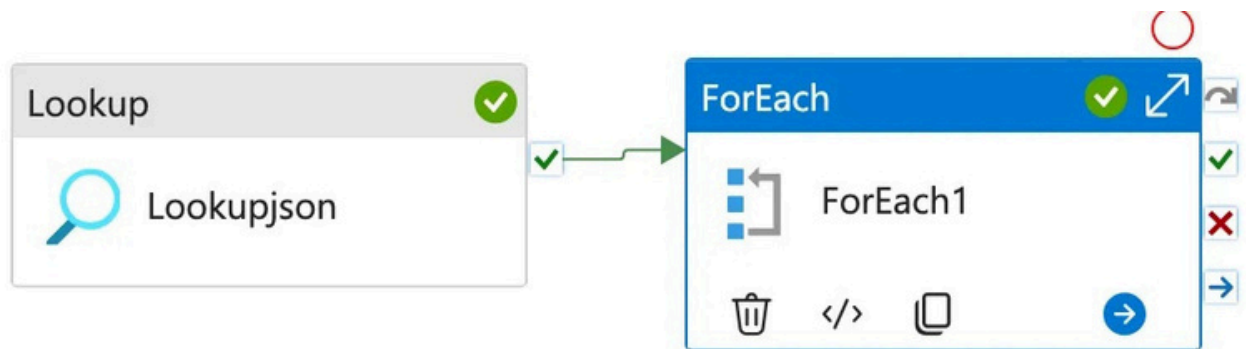
Now, the output of this LookUp Activity will be directed to FarEach Activity. (make sure to uncheck the "First Row" to read the entire data in the json file).

Output

 Copy to clipboard

```
{
  "count": 10,
  "value": [
    {
      "p_rel_url": "anshlamba03/Adventure-Works-Data-Engineering-Project/main/Data/AdventureWorks_Product_Categories.csv",
      "p_sink_folder": "AdventureWorks_Product_Categories",
      "p_sink_file": "AdventureWorks_Product_Categories.csv"
    },
    {
      "p_rel_url": "anshlamba03/Adventure-Works-Data-Engineering-Project/main/Data/AdventureWorks_Calendar.csv",
      "p_sink_folder": "AdventureWorks_Calendar",
      "p_sink_file": "AdventureWorks_Calendar.csv"
    },
    {
      "p_rel_url": "anshlamba03/Adventure-Works-Data-Engineering-Project/main/Data/AdventureWorks_Customers.csv",
      "p_sink_folder": "AdventureWorks_Customers",
      "p_sink_file": "AdventureWorks_Customers.csv"
    }
  ],
}
```

Output of LookUp Activity



General **Settings** Activities (1) User properties

Sequential ☒

Items

@activity('Lookupjson').output.value

I have now connected the Lookupjson Activity to ForEach1 such that the output (all file parameters) will be fed to ForEach Activity (output of Lookup.value).

Inside ForEach Activity canvas, I will paste the Copy Activity which I have created already and input the parameters as following:

General
Source
Sink
Mapping
Settings
User properties

Source dataset *
dynamic_copy_source
Open
New
Preview data

Dataset properties ⓘ

Name	Value
p_rel_url	@item().p_rel_url

Source: Input of p_rel_url from ForEach Item

General
Source
Sink
Mapping
Settings
User properties

Sink dataset *
dynamic_copy_sink
Open
New
Learn more ↗

Dataset properties ⓘ

Name	Value
p_folder_name	@item().p_sink_folder
p_file_name	@item().p_sink_file

Sink: Input of p_folder_name and p_file_name from ForEach Item

Now, I will exit the ForEach Canvas and will run the debug option.

dynamiccopy	✔ Succeeded	Copy data	1/14/2025, 11:31:45 PM 14s
dynamiccopy	✔ Succeeded	Copy data	1/14/2025, 11:31:18 PM 26s
ForEach1	✔ Succeeded	ForEach	1/14/2025, 11:31:18 PM 3m 0s
Lookupjson	✔ Succeeded	Lookup	1/14/2025, 11:31:09 PM 7s

The Pipeline is Executed Successfully

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bronze ...
Container

Search

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: bronze

Search blobs by prefix (case-sensitive)

	Name	Modified	Access tier	Archive status
☐	AdventureWorks_Calendar	14/01/2025, 23:31:57		
☐	AdventureWorks_Customers	14/01/2025, 23:32:15		
☐	AdventureWorks_Product_Categories	14/01/2025, 23:31:33		
☐	AdventureWorks_Products	14/01/2025, 23:32:49		
☐	AdventureWorks>Returns	14/01/2025, 23:33:07		
☐	AdventureWorks_Sales_2015	14/01/2025, 23:33:25		
☐	AdventureWorks_Sales_2016	14/01/2025, 23:33:39		
☐	AdventureWorks_Sales_2017	14/01/2025, 23:33:56		
☐	AdventureWorks_Territories	14/01/2025, 23:34:14		
☐	csvfiles	14/01/2025, 19:16:04		
☐	Product_Subcategories	14/01/2025, 23:32:31		

All the Files are now Uploaded in the bronze container

The BRONZE layer of the Medallion architecture is now completed with a parameterised approach.

THE SILVER LAYER:

Now I will be creating a Databricks Resource under the Resource Group (Azure-Project).

The compute information of the Databricks Cluster is as follows:

Policy ⓘ

Unrestricted

- ☐ Multi node
- ☒ Single node

Access mode ⓘ

No isolation shared

Performance

Databricks Runtime Version

14.3 LTS (includes Apache Spark 3.5.0, Scala 2.12)

- ☒ Use Photon Acceleration ⓘ

Node type ⓘ

Standard_D3_v214 GB Memory, 4 Cores

Compute Information of Cluster

I will now create a notebook (silver_notebook) under a folder (Databricks) inside the Workspace tab.

I will now register an application in Microsoft Entra ID, so that I can connect Databricks to my Storage Account by giving the Credentials of the application.

 Search

 Delete  Endpoints  Preview features

 Overview

 Quickstart

 Integration assistant

 Diagnose and solve problems

> Manage

> Support + Troubleshooting

^ Essentials

Display name : [Azure-Project-App](#)

Application (client) ID : da68acce-1545-4e20-9b7a-863621932b8c

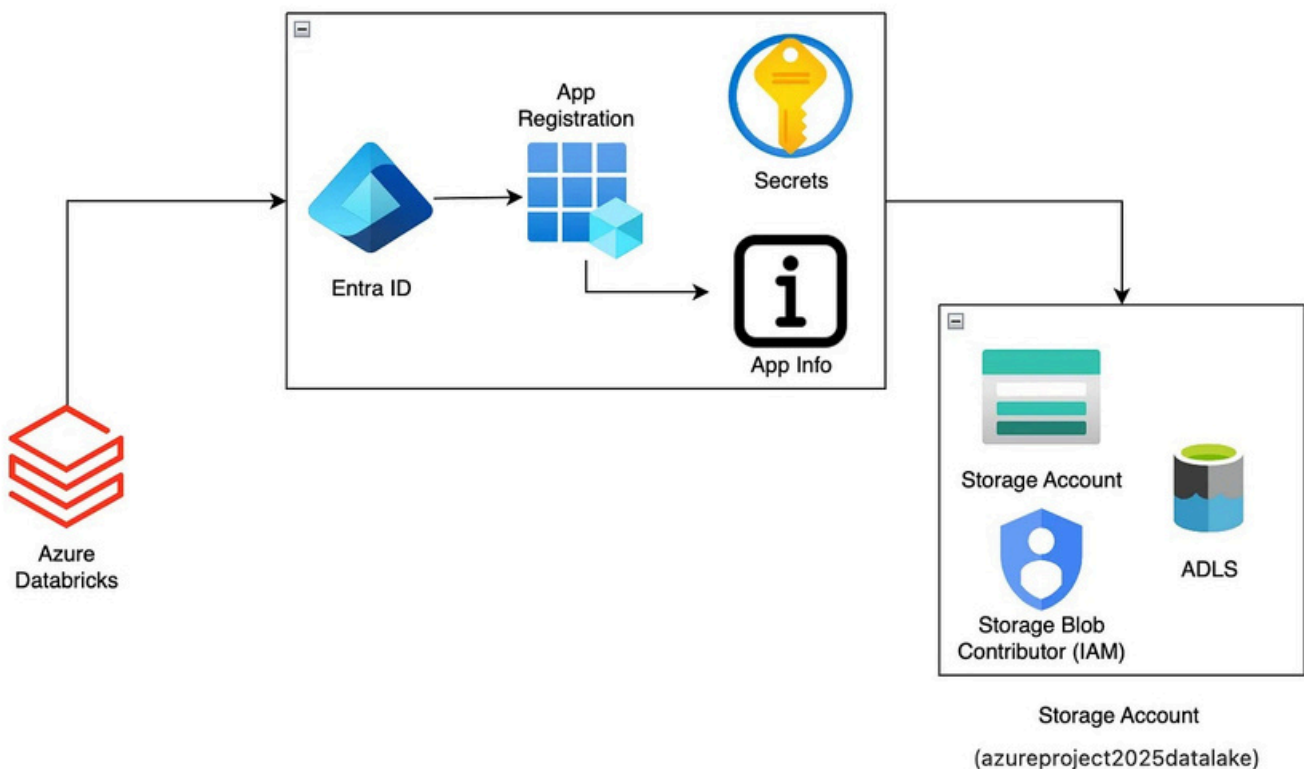
Object ID : ac270846-d487-47d0-acb3-80ea4626cc94

Directory (tenant) ID : 5beb351c-3fb8-418f-b612-fe36ace96ef3

Supported account types : [My organization only](#)

Application Details to connect Databricks and ADLS

Here is a simple flow diagram to connect Azure Databricks with ADLS:



Connection of Azure Databricks and ADLS

I have completed the following steps for connection:

- Copy Paste the information regarding the Application
- Create a secret and Copy the Value of the secret for secure connection
- Grant Storage Blob Data Contributor to Storage Account through IAM and add the Application as the member.

- Apply the connecting Credentials in the Databrick notebook

```
spark.conf.set("fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net", "OAuth")
spark.conf.set("fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.net", "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider")
spark.conf.set("fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net", "<application-id>")
spark.conf.set("fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.net", service_credential)
spark.conf.set("fs.azure.account.oauth2.client.endpoint.<storage-account>.dfs.core.windows.net", "https://login.microsoftonline.com/<directory-id>/oauth2/token")
```

Code for Connecting Azure DataBricks with ADLS

```
df_cal = spark.read.format('csv')\
    .option("header", True)\
    .option("inferSchema", True)\
    .load('abfss://bronze@azureproject2025datalake.dfs.core.windows.net/AdventureWorks_Calendar')

df_cus = spark.read.format('csv')\
    .option("header", True)\
    .option("inferSchema", True)\
    .load('abfss://bronze@azureproject2025datalake.dfs.core.windows.net/AdventureWorks_Customers')
```

Sample Loading of Data

If the code is running properly but you see red underlines in your editor, it likely means there are syntax checking issues within the editor/IDE. These do not affect the execution of the code.

Now I will be performing some Transformations and will push the Transformed data into the silver layer.

PYSPARK TRANSFORMATIONS:


```
05:47 PM (<1s) 7

df_cal = df_cal.withColumn('Month',month(col('Date'))).withColumn('Year',year(col('Date')))

df_cal: pyspark.sql.dataframe.DataFrame = [Date: date, Month: integer ... 1 more field]
```

```
05:47 PM (1s) 8

df_cal.display()
```

(1) Spark Jobs

	Date	Month	Year
1	2015-01-01	1	2015
2	2015-01-02	1	2015

Transformation of df_cal column

I have created a new column called 'Month' and 'Year' by extracting the month and year of the Date column.

withColumn function is used to create or modify a column. It:

Creates a column if we provide different column name.

Modifies a column if we provide same column name.

Now, I will write the df_cal (Transformed data) into the silver container.

```
Just now (1s) 9

df_cal.write.format('parquet')\
  .mode('append')\
  .option("path","abfss://silver@azureproject2025datalake.dfs.core.windows.net/AdventureWorks_Calendar")\
  .save()

(1) Spark Jobs
```

Load the data to silver container



Search

Upload Add Directory Refresh Rename Delete Change tier

Overview

Diagnose and solve problems

Access Control (IAM)

> Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: silver / AdventureWorks_Calendar

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ [..]		
☐ _committed_9111310690737742703	15/01/2025, 13:38:51	Hot (Inferred)
☐ _started_9111310690737742703	15/01/2025, 13:38:50	Hot (Inferred)
☐ _SUCCESS	15/01/2025, 13:38:51	Hot (Inferred)
☐ part-00000-tid-9111310690737742703...	15/01/2025, 13:38:51	Hot (Inferred)

Data Successfully Loaded into the silver container

Now I will be performing Transformations on df_cus data.

Just now (<1s) 11

```
df_cus = df_cus.withColumn('FullName', concat(col('Prefix'), lit(' '), col('FirstName'), lit(' '), col('LastName')))
```

df_cus: pyspark.sql.dataframe.DataFrame = [CustomerKey: integer, Prefix: string ... 12 more fields]

Just now (<1s) 12

```
df_cus = df_cus.drop('Prefix', 'FirstName', 'LastName')
```

df_cus: pyspark.sql.dataframe.DataFrame = [CustomerKey: integer, BirthDate: date ... 9 more fields]

Just now (1s) 13

```
df_cus.display()
```

(1) Spark Jobs

	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	FullName
1	\$90,000	2	Bachelors	Professional	Y	MR. JON YANG
2	\$60,000	3	Bachelors	Professional	N	MR. EUGENE HUANG

Creating FullName Column using concat() function

Here I have used the following functions:

concat: concatenates different columns

lit: since (' ') is a space (i.e a constant) I have used lit function.

- drop: I have dropped prefix, first and last name because I have combined them into a single column.

I will now write df_cus data into silver container and the code is same as df_cal except that the dataframe name only must be changed.

Now, I will Transform df_pro data.

▶

✓ Just now (<1s)

17

df_pro.display()

▶ (1) Spark Jobs

Table ▾ +

🔍

🔼

📄

	1 ² ₃ ProductKey	1 ² ₃ ProductSubcategoryKey	A ^B _C ProductSKU	A ^B _C ProductName	A ^B _C ModelName	A ^B _C ProductDes
1	214	31	HL-U509-R	Sport-100 Helmet, Red	Sport-100	Universal fit, we
2	215	31	HL-U509	Sport-100 Helmet, Black	Sport-100	Universal fit, we
3	218	23	SO-B909-M	Mountain Bike Socks, M	Mountain Bike Socks	Combination of
4	219	23	SO-B909-L	Mountain Bike Socks, L	Mountain Bike Socks	Combination of

Sample Snippet of df_pro Data

I will perform a Transformation that retrieves the size of a product from ProductSKU.

Notice that some of the records in ProductSKU doesn't have any size to it. In that case we will get NULL values. I will try to fill the NULL with the value Not Applicable.

1 minute ago (<1s) 18

```
df_pro = df_pro.withColumn('ProductSize', split(col('ProductSKU'), '-')[2])
```

df_pro: pyspark.sql.dataframe.DataFrame = [ProductKey: integer, ProductSubcategoryKey: integer ... 9 more fields]

Just now (<1s) 19

```
df_pro = df_pro.fillna({'ProductSize': 'Not Applicable'})
```

df_pro: pyspark.sql.dataframe.DataFrame = [ProductKey: integer, ProductSubcategoryKey: integer ... 9 more fields]

Just now (<1s) 20

```
df_pro.display()
```

(1) Spark Jobs

		A _C ^B ProductColor	A _C ^B ProductSize	A _C ^B ProductStyle	1.2 ProductCost	1.2 ProductPrice
1		Red	R	0	13.0863	34.99
2		Black	Not Applicable	0	12.0278	33.6442
3		White	M	U	3.3963	9.5
4		White	L	U	3.3963	9.5
5		Blue	B	0	12.0278	33.6442
6		Multi	Not Applicable	U	5.7052	8.6442

Created ProductSize and filled NULL with Not Applicable

I will now write the df_pro into the silver container.

I will analyse data in df_sales using some Transformations.

Just now (1s) 26

```
df_sales.groupby(col('TerritoryKey')).agg(sum(col('OrderQuantity')).alias('TotalQuantity')).display()
```

(2) Spark Jobs

	1 ₃ ² TerritoryKey	1 ₃ ² TotalQuantity
1	1	12513
2	6	10894
3	3	30
4	5	49
5	9	17951

GroupBy Function

GroupBy Aggregation:

- I grouped the data in accordance with TerritoryKey to see the performance of Orders based on Territories.

- I have performed a sum aggregation on OrderQuantity column to see the total number of orders.
- Finally I have used alias function to rename the sum(OrderQuantity) column.

▶

✓ 06:06 PM (<1s)

27

```
df_sales = df_sales.withColumn('OrderDate', to_timestamp(col('OrderDate')))
```

▶

df_sales: pyspark.sql.dataframe.DataFrame = [OrderDate: timestamp, StockDate: date ... 6 more fields]

▶

✓ 06:08 PM (<1s)

28

```
df_sales = df_sales.withColumn('OrderNumber', regexp_replace(col('OrderNumber'), 'SO', 'ON'))
```

▶

df_sales: pyspark.sql.dataframe.DataFrame = [OrderDate: timestamp, StockDate: date ... 6 more fields]

▶

✓ 06:09 PM (<1s)

29

```
df_sales = df_sales.withColumn('OrderLineColumn', col('OrderLineItem') * col('OrderQuantity'))
```

▶

df_sales: pyspark.sql.dataframe.DataFrame = [OrderDate: timestamp, StockDate: date ... 7 more fields]

▶

✓ Just now (1s)

30

```
df_sales.select(col('OrderLineColumn'), col('OrderDate'), col('OrderNumber')).display()
```

▶ (1) Spark Jobs

Table ▾ +

	1 ² 3 OrderLineColumn	📅 OrderDate	A ^B C OrderNumber
1	4	2017-01-01T00:00:00.000+00:00	ON61285
2	3	2017-01-01T00:00:00.000+00:00	ON61285
3	1	2017-01-01T00:00:00.000+00:00	ON61285

I have made the following Transformations to df_sales data:

- regexp_replace function replaces a string with the string we want. In this case I have replaces SO with ON (Order Number).
- Performed arithmetic operation on columns where OrderLineColumn is obtained by multiplying OrderLineItem and OrderQuantity.
- Sometimes downstream professionals require date information in timestamps.
- So, I have converted the OrderDate into timestamp.

- Finally, I have used Select function to show only the changed and modified columns instead of the whole data frame.

Now, I will write all the remaining data frames to silver container.

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silver ...
Container

Search

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

Authentication method: Access key ([Switch to Microsoft Entra user account](#))
Location: silver

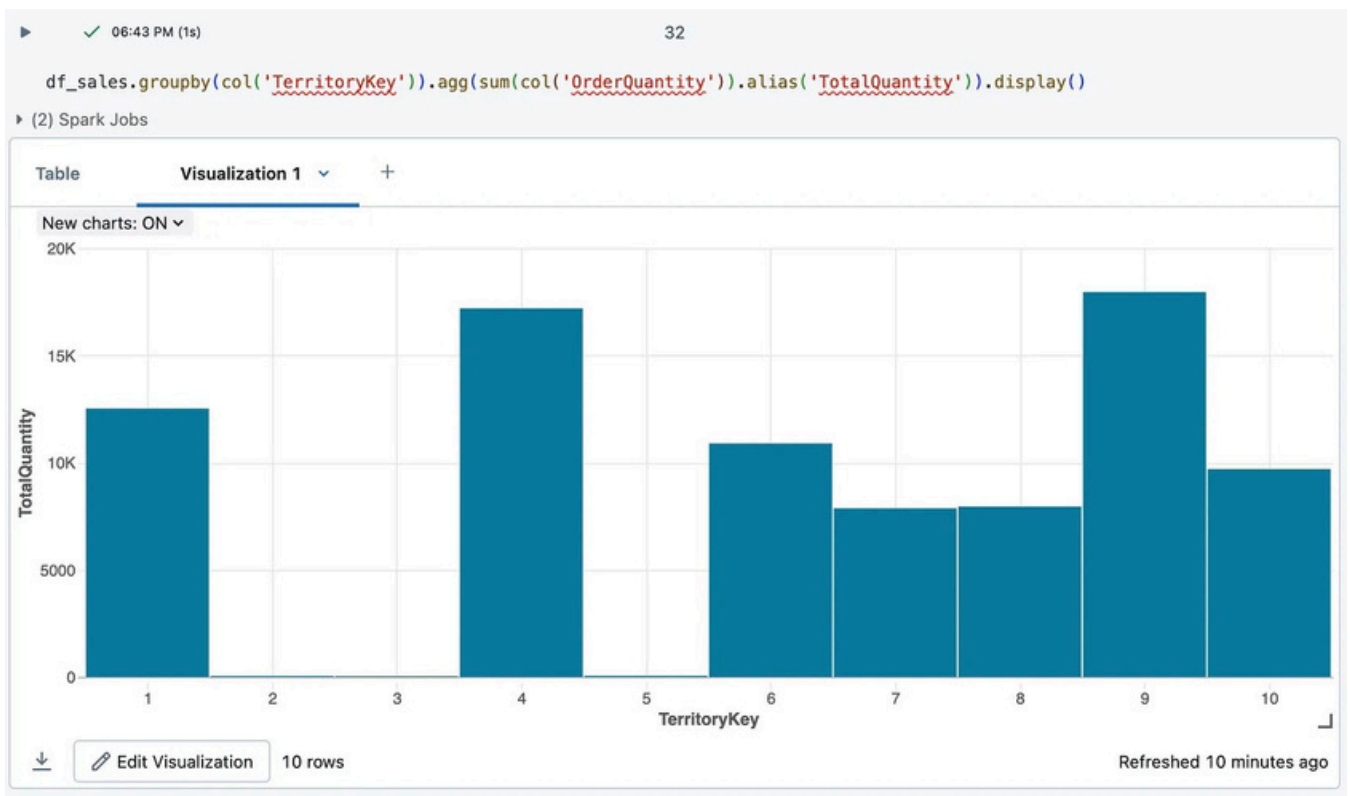
Search blobs by prefix (case-sensitive)

	Name	Modified	Access tier	Archive status
☐	AdventureWorks_Calendar	15/01/2025, 17:47:57		
☐	AdventureWorks_Customers	15/01/2025, 17:48:17		
☐	AdventureWorks_ProductCategory	15/01/2025, 17:51:24		
☐	AdventureWorks_Products	15/01/2025, 17:48:42		
☐	AdventureWorks>Returns	15/01/2025, 17:49:41		
☐	AdventureWorks_Sales	15/01/2025, 18:28:31		
☐	AdventureWorks_SubCat	15/01/2025, 17:48:21		
☐	AdventureWorks_Territories	15/01/2025, 17:49:40		

Transformed data in silver container

VISUALISATION IN DATABRICKS:

Scenario 1:



Territory-wise Performance

I have done a Territory-wise analysis where we can understand which territory is best performing in sales.

Scenario 2:



Month-wise and Year-wise Sales Analysis

Here I have performed groupby on both OrderYear and OrderMonth such that the stakeholders can easily hover over a specific month and see the total quantity sold by the distributors.

SCENARIO 3:



Here, we can see that most of the Stock manufacturing in the year 2004 were did in first few months and the Stock manufacturing in the year 2003 were did in last months of the year.

The SILVER layer of the Medallion architecture is now completed where all the Transformation and Visualisation is done in Databricks and is written into the silver container.

For the gold layer presentation, I will be using Azure Synapse Analytics.

Create Synapse workspace ...

Create a Synapse workspace to develop an enterprise analytics solution in just a few clicks.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.

Subscription * ⓘ	Free Trial	▼
Resource group * ⓘ	Azure-Project	▼
	Create new	
Managed resource group ⓘ	synapse-managed-group	✓

Workspace details

Name your workspace, select a location, and choose a primary Data Lake Storage Gen2 file system to serve as the default location for logs and job output.

Workspace name *	synapse-project-workspace-2025	✓
Region *	East US	▼
Select Data Lake Storage Gen2 * ⓘ	☒ From subscription ☐ Manually via URL	
Account name * ⓘ	(New) defaultstorage2025	▼
	Create new	
File system name *	(New) defaultfile2025	▼
	Create new	

Creation of Synapse Workspace

I have created a default Storage Account and default File System which will be using by Synapse Analytics.

Create Synapse workspace ...

* Basics * **Security** Networking Tags Review + create

Configure security options for your workspace.

Authentication

Choose the authentication method for access to workspace resources such as SQL pools. The authentication method can be changed later on. [Learn more](#)

Authentication method ⓘ

- ☒ Use both local and Microsoft Entra ID authentication
☐ Use only Microsoft Entra ID authentication

SQL Server admin login * ⓘ

adminsuj ✓

SQL Password ⓘ

..... ✓

Confirm password

..... ✓

System assigned managed identity permission

Select to grant the workspace network access to the Data Lake Storage Gen2 account using the workspace system identity. [Learn more](#)

☐ Allow network access to Data Lake Storage Gen2 account. ⓘ

i The selected Data Lake Storage Gen2 account does not restrict network access using any network access rules, or you selected a storage account manually via URL under Basics tab. [Learn more](#)

Creation of SQL Server Login

Add role assignment ...

Role **Members** Conditions Review + assign

Selected role Storage Blob Data Contributor

Assign access to
☐ User, group, or service principal
☒ Managed identity

Members + Select members

Name	Object ID	Type	
azure-workspace-2025	31d792d5-40e8-47e7-8fad-05a5722c3a...	Synapse workspace ⓘ	🗑️

Description

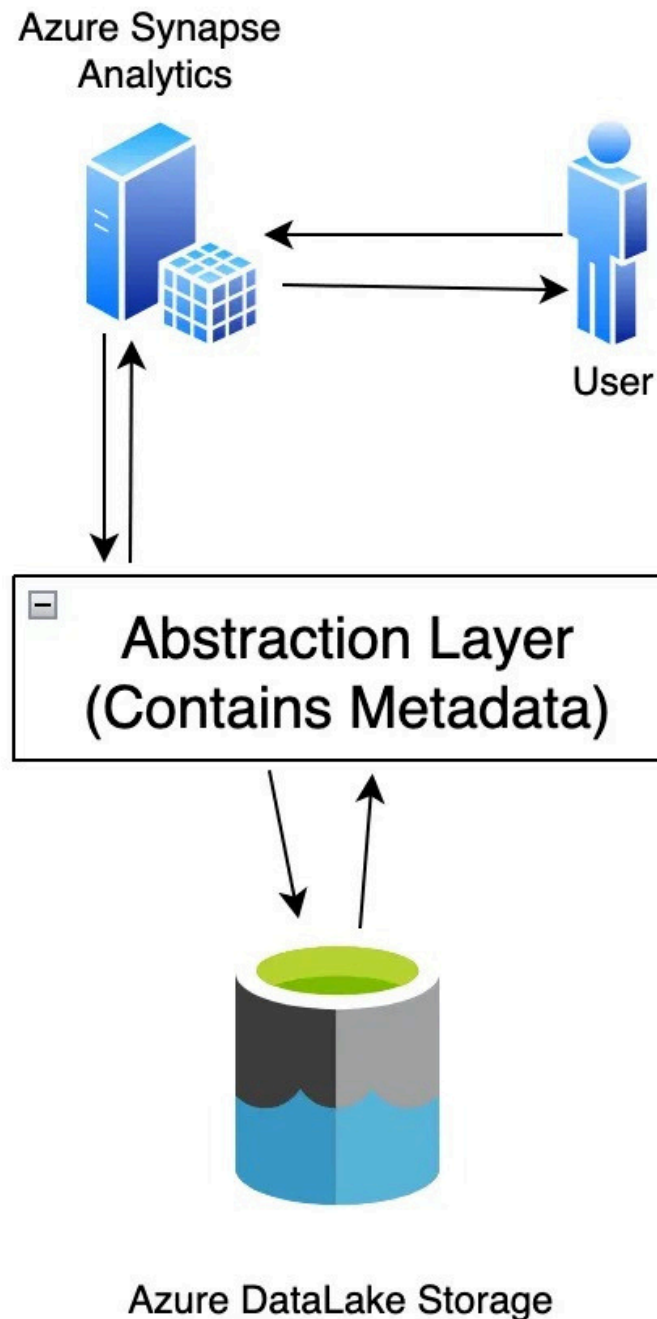
Allowed Storage Blob Data Contributor using Managed identity and assigned, Azure Synapse Workspace as the member.

Creation of Managed Identity

Now, I have created a managed identity under the Storage blob data contributor.

Managed Identity helps in linking of Azure Resources with each other. Now Synapse Analytics can access data from ADLS.

Azure Synapse Analytics helps us in implementing lakehouse concept.



The above diagram represents an abstraction layer over Azure Data Lake Storage, allowing SQL Server and users to query data using metadata.

Synapse Analytics retrieves data from Azure Data Lake Storage (ADLS) and enables the implementation of SQL queries, advanced analytics, and other functionalities, making it easier to process, analyse, and visualise large-scale data efficiently.

Add role assignment

Role **Members** Conditions Review + assign**Selected role** Storage Blob Data Contributor**Assign access to** ☒ User, group, or service principal
☐ Managed identity**Members** [+ Select members](#)

Name	Object ID	Type	
S Suraj Badrinarayanan	20698592-c7d1-4a47-839a-fb570ffa0d3	User	

Description

Optional

IAM to Query the Data as a User

```
14 SELECT * FROM
15     OPENROWSET (
16         BULK 'https://azureproject2025datalake.dfs.core.windows.net/silver/AdventureWorks\_Calendar/',
17         FORMAT = 'PARQUET'
18     ) as query1
19
```

Results MessagesView **Table** Chart [Export results](#)  Search

Date	Month	Year
2015-01-01	1	2015
2015-01-02	1	2015
2015-01-03	1	2015
2015-01-04	1	2015

I have now queried the data in ADLS, using SQL Syntax and displayed the data in parquet file format in the form of a table.

OPENROWSET() function created an abstraction layer on top of ADLS and made me perform SQL Queries on parquet file format.

Now, I will create a gold schema where I will store all the Views inside it.

```

1  -- CREATE VIEW CALENDAR
2  CREATE VIEW gold.calendar AS
3  SELECT
4      *
5  FROM
6      OPENROWSET(
7          BULK 'https://azureproject2025datalake.blob.core.windows.net/silver/AdventureWorks\_Calendar/',
8          FORMAT = 'PARQUET'
9      ) AS QUER1;
10
11 -- CREATE VIEW CUSTOMERS
12 CREATE VIEW gold.customers AS
13 SELECT
14     *
15 FROM
16     OPENROWSET(
17         BULK 'https://azureproject2025datalake.blob.core.windows.net/silver/AdventureWorks\_Customers/',
18         FORMAT = 'PARQUET'
19     ) AS QUER1;

```

Successfully Created Views for all Data in the silver Layer

Now the stakeholders, managers or data analysts can query the data as if it was a SQL Server, but in reality, all the data are retrieved from ADLS using an Abstraction Layer.

Now, I have successfully created the gold schema in Azure Synapse Analytics and completed the requirements in accordance with the Medallion Architecture.

Azure Synapse Analytics simplifies data processing by allowing you to query large datasets directly from Azure Data Lake without the need for expensive SQL databases. Synapse makes it easy to run SQL queries on big data, providing fast insights while keeping expenses low, making it a powerful tool for modern businesses.

I hope you found this guide helpful! If you enjoyed the blog or have any thoughts to share, feel free to leave a like or drop a comment on my Medium blog — I'd love to hear your feedback!